

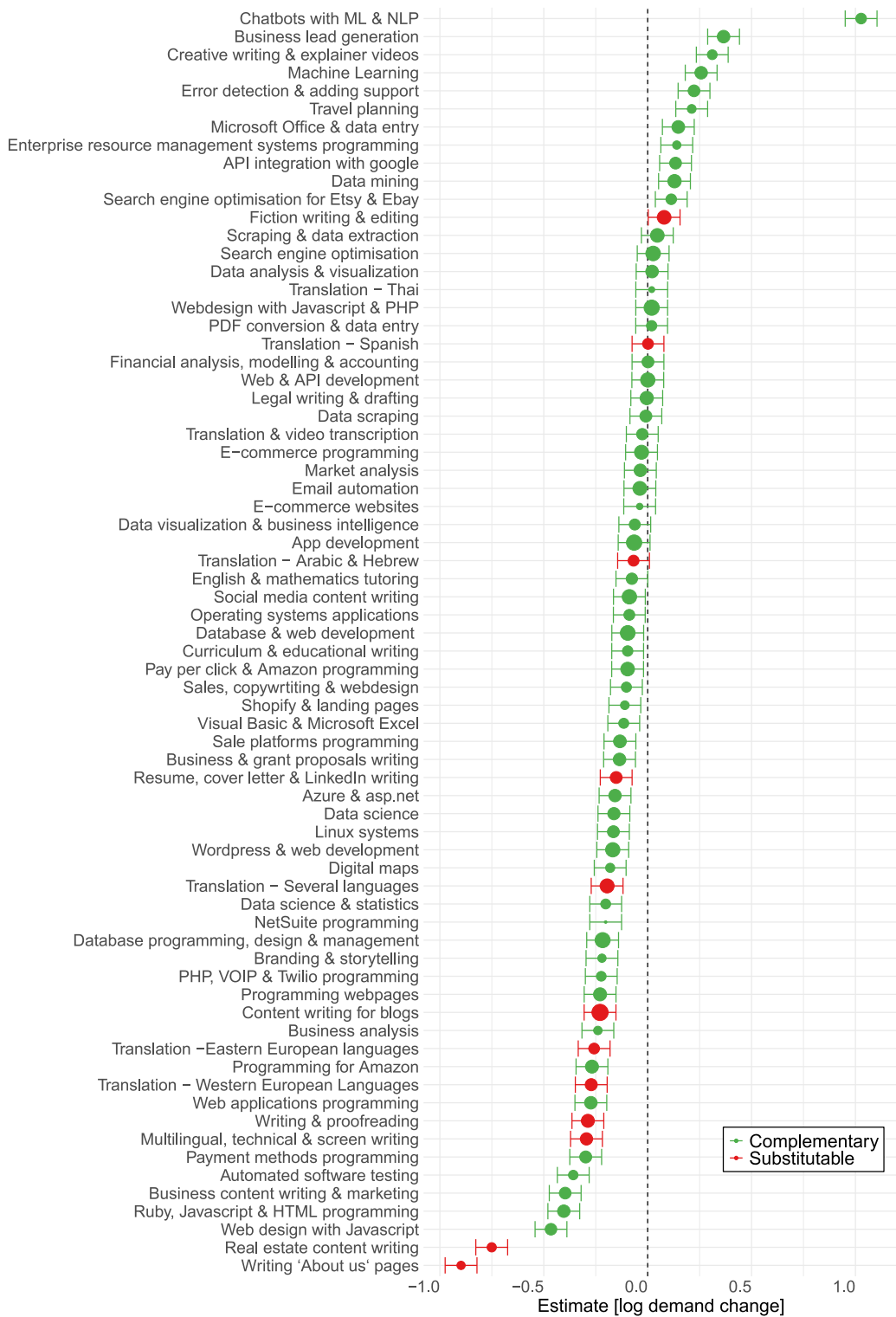


Fig. 5. Estimates of log change in demand at the skill cluster level. Error bars denote standard errors, and the node size is proportional to the log average number of job postings in the given skill cluster.

Randomization inference. As noted by MacKinnon and Webb (2020), randomization inference can provide more accurate p-values when the share of treated clusters is small relative to the control clusters, resulting in either too large or too small standard errors.

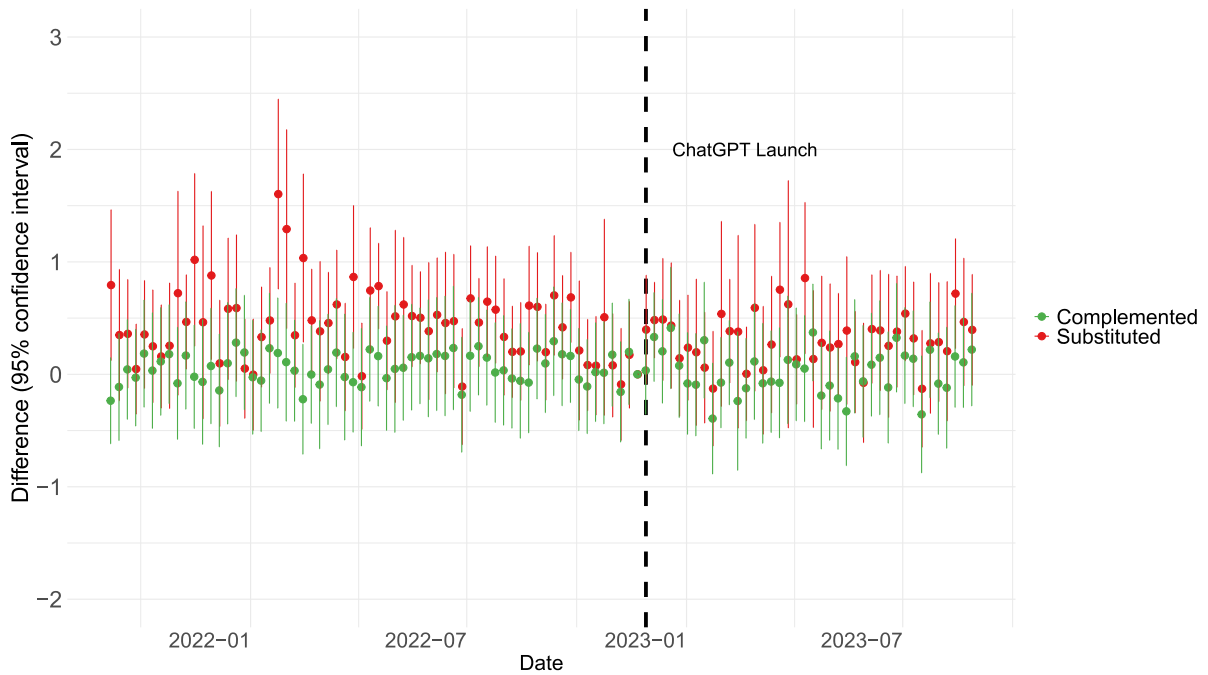


Fig. 6. Event study: log mean budget - This figure plots the difference in log mean budget between the unaffected cluster and substituted and augmented clusters and the corresponding 95% confidence intervals. The vertical line corresponds to ChatGPT launch. Standard errors are clustered at job posting cluster level.

Accordingly, we implemented a randomization inference procedure as follows: we first shuffled the treatment labels (Unaffected, Complementary, Substitutable) randomly. Then, we re-estimated the difference-in-differences model given by Eq. (1) using the shuffled treatment labels, saving the difference-in-differences terms δ_1 and δ_2 . Repeating this procedure many times produces a distribution of δ_1 and δ_2 pairs under the null hypothesis that the launch of ChatGPT has no effect on the demand for complementary and substitutable clusters. Comparing the actual estimates of δ_1 and δ_2 to this distribution of placebo estimates allows us to understand how extreme they are in comparison. We report the results of this exercise in Appendix H. The results indicate that the p-values based on randomization inference are very similar to those based on the cluster robust covariance matrix reported in the main text.

3.2.5. Isolating the demand shock from general equilibrium effects

The previous analysis demonstrated a reduction in job postings for substitutable skill clusters following the launch of ChatGPT. Here, we strengthen the argument that this decline reflects a demand-side shock rather than a reduction in labor supply with three key points.

First, we highlight, that using *job postings* as a measure of labor demand is a well-established approach in labor economics (Davis et al., 2013; Marinescu and Wothoff, 2020; Barnichon, 2010). Job postings are initiated by employers to signal a need for labor, making them a relatively pure indicator of demand, independent of supply-side factors such as the number or characteristics of job seekers.

Second, we show that the mean budget per job posting spent by employers require complementary and substituted skill clusters did not change due to the introduction of ChatGPT. This suggests that employers were willing to pay the same price for labor but demanded less of it, consistent with a leftward shift in the labor demand curve following the ChatGPT launch. Although we lack data on realized wages for all employers, some job postings include a desired employer budget as a part of the job posting. We use the subset data of job postings of employers who reported a budget, and replicate the event study analysis of the previous section, but with the log mean budget as the dependent variable. Fig. 6 and Column (1) of Table 3 show the results of this analysis. There is no significant change in the average employer budget across complementary and substitutable clusters in comparison to the unaffected category.

Third, we show that applications per job posting increased due to a decrease in job postings rather than an increase in applications. To do this, we replicate the event study analysis using the log of the average number of applicants per job posting as the dependent variable. As Fig. 7 and Column (2) of Table 3 show, the number of applicants per project increased after the launch of ChatGPT. Moreover, we can decompose the increase in applications per job posting as follows

$$\log\left(\frac{\sum_j Applicants_{jit}}{Postings_{it}}\right) = \log\left(\sum_j Applicants_{jit}\right) - \log(Postings_{it}). \quad (3)$$

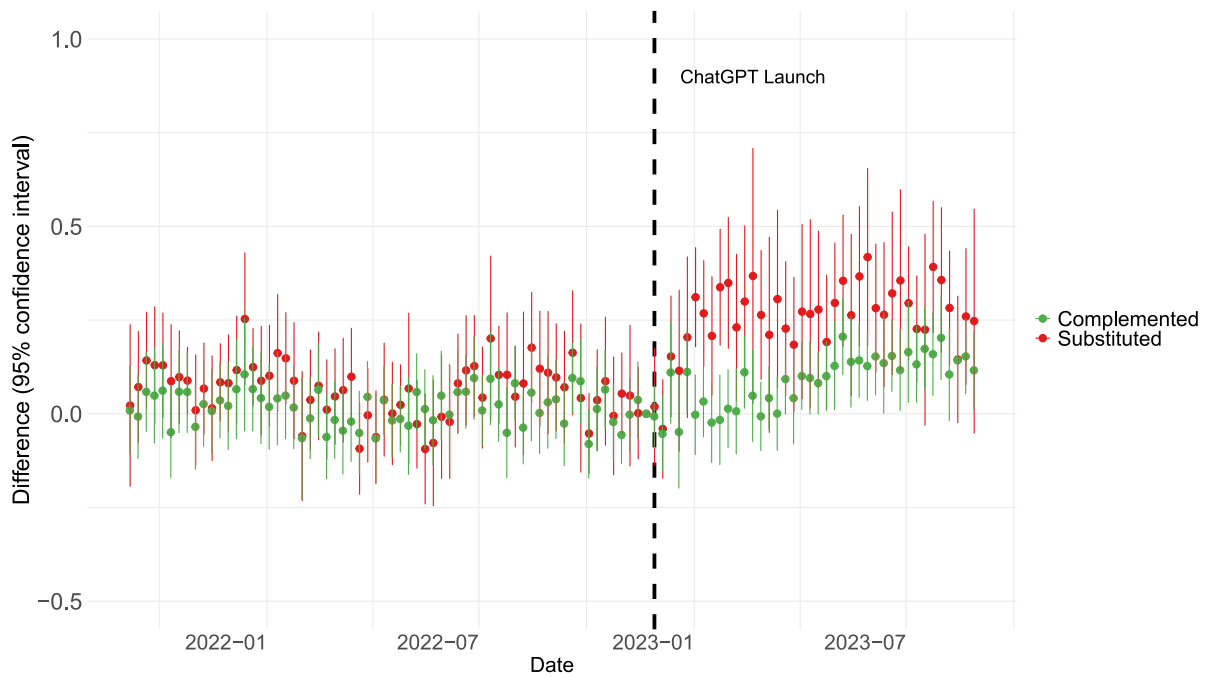


Fig. 7. Event study: log applicants per job posting - This figure plots the difference in log mean number of applicants between the unaffected cluster and substituted and augmented clusters and the corresponding 95% confidence intervals. The vertical line corresponds to ChatGPT launch. Standard errors are clustered at job posting cluster level.

Table 3

Difference-in-Difference Results - Additional Outcomes: Column (1) reports the difference-in-difference estimates for the effect of ChatGPT on employers' reported budgets (measured in log-\$) across the two treatment groups compared to the control group. Column (2) reports the effect of ChatGPT launch on average applicants per job posting (log) across the two treatment groups compared to the control group. Standard errors are clustered at the cluster level.

	(1)	(2)
Dependent Variable	log(Avg. budget)	log(Avg. Applicants/Job posting)
Intercept	6.44*** (0.13)	2.97*** (0.05)
Substituted	-0.50** (0.18)	0.05 (0.10)
Complemented	-0.02 (0.15)	0.04 (0.08)
After	0.08* (0.03)	0.06*** (0.02)
Substituted × After	-0.13 (0.07)	0.20*** (0.05)
Complemented × After	-0.03 (0.04)	0.07** (0.03)
Observations	11 930	11 957
R ²	0.02	0.03

Notes: *p < 0.05, **p < 0.01, ***p < 0.001.

From Column (1) of Table 2 and Column (2) of Table 3 we know that $\log(Postings_{it}) = -0.28$ and $\log\left(\frac{\sum_j Applicants_{jit}}{Postings_{it}}\right) = 0.20$, with respective standard errors 0.09 and 0.05. The difference between the estimates is within the standard errors, and therefore we cannot attribute the increase in applications per job posting to an increase in the supply.⁶

⁶ More formally: let $R = \log\left(\frac{\sum_j Applicants_{jit}}{Postings_{it}}\right)$, $A = \log\left(\sum_j Applicants_{jit}\right)$, and $P = \log(Postings_{it})$. The empirical estimates imply, that $\Delta R = 0.2$, and $\delta P = -0.28$. Solving for ΔA yields -0.08 . To assess the statistical significance of ΔA , we can solve for its standard error: $SE(\Delta A) = \sqrt{SE(\Delta R)^2 + SE(\delta P)^2} \approx .10$ (assuming independence between R and P).

In summary, we have demonstrated that the launch of ChatGPT led to a decrease in labor demand without reducing average employer budgets. Furthermore, the number of applicants per job posting increased after the launch, and this increase can be almost completely explained by the reduction in the number of job postings. Taken together, these findings suggest that the launch of ChatGPT led to a reduction in labor demand, with the same number of applicants competing for a smaller pool of substitutable jobs.

While our data does not capture wage changes, it is possible that increased competition may have reduced equilibrium wages. If this is the case, our demand estimates should be interpreted as a lower bound on the true effect. However, Liu et al. (2023), using a similar dataset and research design, did not find evidence of reduced project values, suggesting that the impact on workers' wages may be limited.

One limitation of the previous analysis is that we lack information about the identity or characteristics of the applicants, which may have changed over time. If the launch of ChatGPT led to a less qualified applicant pool, this could have also contributed to a reduction in labor demand. This shift in applicant quality — along with the possibility of ChatGPT directly replacing certain types of work — could be one reason for the observed decrease in labor demand following the launch of ChatGPT. However, we think this is unlikely for two reasons. First, we do not find an increase in supply, meaning that if there was an influx of less qualified applicants, more qualified workers must have left the platform. We have no reason to suspect this would have happened given that the average budget per job remained constant. Moreover, the platform in question allows employers to closely monitor workers' performance and even withhold payment if the deliverable is deemed insufficient in quality. This level of oversight reduces the likelihood that employers would decrease labor demand purely due to uncertainty about worker quality.

4. Discussion

To understand how AI is transforming labor markets, it is crucial to understand its impact on demand across diverse skill sets, distinguishing between jobs that AI can substitute and those it can complement. In this paper, we analyze job posting data from online labor markets, which are highly flexible and quick to react to external changes. We use transformer-based topic modeling to cluster over 3 million job postings into 116 fine-grained skill clusters. Additionally, we employ prompt engineering to label these clusters as substitutable by, complementary to, or unaffected by AI. Finally, we employ a difference in differences analysis to study the impact of ChatGPT's release on demand across 71 skill clusters which we have categorized as either substitutable by or complementary to AI.

Following the introduction of ChatGPT, we find a reduction in demand of approximately 24% for skills deemed substitutable by LLMs compared to unaffected skill clusters, illustrating the potential of LLMs to automate certain tasks historically performed by humans. Conversely, the overall demand for skills complementary to LLMs did not change significantly. However, some complementary skill clusters, such as those related to chatbot development and machine learning, showed substantial increases in demand, while others experienced slight decreases. Regarding wages and geography, substitutable skills are associated with higher wages and are typically performed by freelancers in high-wage countries. In contrast, complementary skills exhibit a broader wage spectrum and are often performed by freelancers from middle-income countries. The impact varies by project duration and worker experience levels, with demand decreasing mainly in short-term projects within the substitutable category. Moreover, while complementary skill clusters did not see a reduction in demand as a whole, novice workers within complementary skill clusters experienced a significant contraction.

Our results provide a nuanced perspective on the labor market effects of generative AI technologies like ChatGPT and other LLMs. Unlike previous literature, which focused primarily on a limited subset of skill clusters in the online freelancing market (Hui et al., 2023; Demirci et al., 2023; Qiao et al., 2023; Liu et al., 2023), our analysis reveals both job destruction in the substitution category and job creation in certain complementary skill clusters. Within the writing and translation sectors, our study uncovers significant heterogeneity: for instance, demand for “about us” page writing has decreased more than for multilingual and technical writing, while translation demand is most affected for Western European languages. Although the decline in demand for substitutable work is concerning, it is predominantly concentrated in short-term gigs rather than long-term projects. Additionally, we highlight the role of new technologies in driving job creation, exemplified by the increased demand for AI-powered chatbots. The advancements in LLMs have significantly improved chatbot quality, significantly expanding their possible use cases and consequently driving higher demand for chatbot development.

The decrease in demand for novice workers within complementary skill clusters may seem to contradict experimental evidence suggesting that AI tools benefit inexperienced workers the most (Noy and Zhang, 2023; Brynjolfsson et al., 2023). However, while these experimental studies are conducted in controlled environments where workers are directly given the AI tools, in the real labor market, it is typically companies or clients who outsource work to freelancers. Thus, novice workers within firms may benefit from AI tools, reducing the need to hire external freelancers. This underscores the fact that the impact of technologies on workers depends on how tools are implemented and who is provided access to them.

This paper's analysis has limitations. First, online labor platforms evolve over time, introducing new features and disabling others, which introduces a degree of noise into our data collection. As outlined in Section 2.3, skill names change over time (e.g., *Photoshop* to *Adobe Photoshop*), and as discussed in Section 2.2, we have occasional gaps in our data due to software errors and platform interface changes. However, using transformer-based hierarchical topic modeling, we create a stable time series of skill clusters. Our results remained consistent when controlling for or excluding the few problematic periods, indicating that the missing data adds only random noise rather than systematic bias. Another limitation is our reliance on GPT-4 for labeling skill clusters exposure to automation. This is a highly challenging task, even for expert human labelers, as it requires specialized knowledge